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STROKE PATIENT TRANSPORT PROTOCOL

Prehospital and Interfacility Transport — EMS & Stroke System of Care

Document Control

- Document Number: SU-002-PROT-03
- Title: Stroke Patient Transport Protocol
- Effective Date: [To be assigned upon approval]
- Review Date: Annually
- Approved By: Stroke Committee / EMS Medical Director

1. Purpose

To ensure timely, safe, and appropriate transport of patients with suspected acute stroke to the most appropriate stroke center while minimizing delays to reperfusion therapy.

2. Scope

This protocol applies to:

- EMS agencies
- Emergency Department staff
- Stroke Team
- Transfer Center
- Critical Care Transport Teams
- Air Medical Transport (when applicable)

3. Transport Eligibility

Patients shall be transported under the Stroke Alert pathway if they have:

- Sudden onset of focal neurological deficit
- Positive validated stroke screening tool (FAST, BE-FAST, CPSS, etc.)
- Last Known Well (LKW) documented or estimated
- Symptoms suggestive of acute ischemic or hemorrhagic stroke

4. Destination Selection

A. Primary Stroke Center (PSC)

Transport to the nearest PSC when:

- Suspected acute stroke without evidence of large vessel occlusion (LVO)

- Estimated transport time to Comprehensive Stroke Center (CSC) would cause significant treatment delay
- Patient is a potential candidate for intravenous thrombolysis

B. Comprehensive Stroke Center (CSC)

Transport directly to a CSC when:

- Positive LVO screening (FAST-ED, RACE, LAMS, VAN, etc.)
- Suspected need for mechanical thrombectomy
- Severe stroke (e.g., NIHSS ≥ 6, according to regional protocol)
- Intracerebral hemorrhage requiring neurosurgical care
- Regional EMS destination criteria are met

5. Transport Priorities

EMS should:

- Minimize scene time (target ≤ 15 minutes)
- Obtain IV access if it does not delay transport
- Check blood glucose
- Monitor airway, breathing, circulation, and neurological status
- Avoid unnecessary procedures that delay arrival

6. Hospital Pre-Notification

EMS shall notify the receiving hospital before arrival and provide:

- Estimated time of arrival (ETA)
- Patient age and sex
- Last Known Well
- Stroke screening result
- LVO score (if performed)
- Blood glucose
- Blood pressure
- Current neurological deficits
- Anticoagulant use (if known)
- Airway status

7. Interfacility Transfer Protocol

Patients requiring higher-level stroke care shall be transferred without unnecessary delay.

Indications Include

- Mechanical thrombectomy
- Neurosurgical intervention
- Neurocritical care
- Advanced neuroimaging unavailable at referring hospital
- Complex stroke management beyond local capability

Prior to Transfer, the Referring Hospital Shall

- Stabilize airway, breathing, and circulation
- Perform CT brain (and CTA if available)
- Send imaging electronically whenever possible
- Complete transfer documentation
- Notify the receiving stroke team
- Arrange the fastest appropriate transport

8. Mode of Transport

Ground Ambulance

Preferred when transport time is acceptable and rapid access is available.

Air Ambulance

Consider when:

- Long transport distance
- Time-sensitive thrombectomy candidate
- Remote location
- Ground transport would significantly delay definitive treatment

9. Monitoring During Transport

Monitor and document:

- Blood pressure
- Heart rate
- Respiratory rate
- Oxygen saturation
- Glasgow Coma Scale (GCS)
- Neurological status
- Blood glucose (if indicated)

Maintain

- $SpO_2 \geq 94\%$
- Normoglycemia
- Airway protection
- Intravenous access

10. Documentation

Record:

- Symptom onset / Last Known Well
- Stroke screening tool results
- LVO assessment
- Vital signs
- Blood glucose
- Treatments provided
- Time of EMS activation
- Departure time
- Arrival time
- Hospital prenotification time
- Any clinical deterioration during transport

11. Quality Indicators

Indicator	Target
EMS prenotification	≥ 95%
Last Known Well documented	≥ 95%
Blood glucose documented	≥ 95%
Scene time	≤ 15 minutes
Appropriate destination compliance	≥ 90%
Door-In–Door-Out (transfer hospitals)	≤ 60 minutes (goal)
Transfer imaging available before departure	≥ 95%

12. Staff Education

All EMS and transport personnel shall receive annual education on:

- Stroke recognition
- Stroke screening tools
- LVO assessment
- Destination decision-making
- Safe transport practices
- Hospital prenotification
- Documentation requirements

Competency should be assessed annually.

References

- American Heart Association / American Stroke Association. Guidelines for the Early Management of Patients With Acute Ischemic Stroke.
- Regional EMS Stroke Destination Policy.
- Hospital Stroke Program Policies.